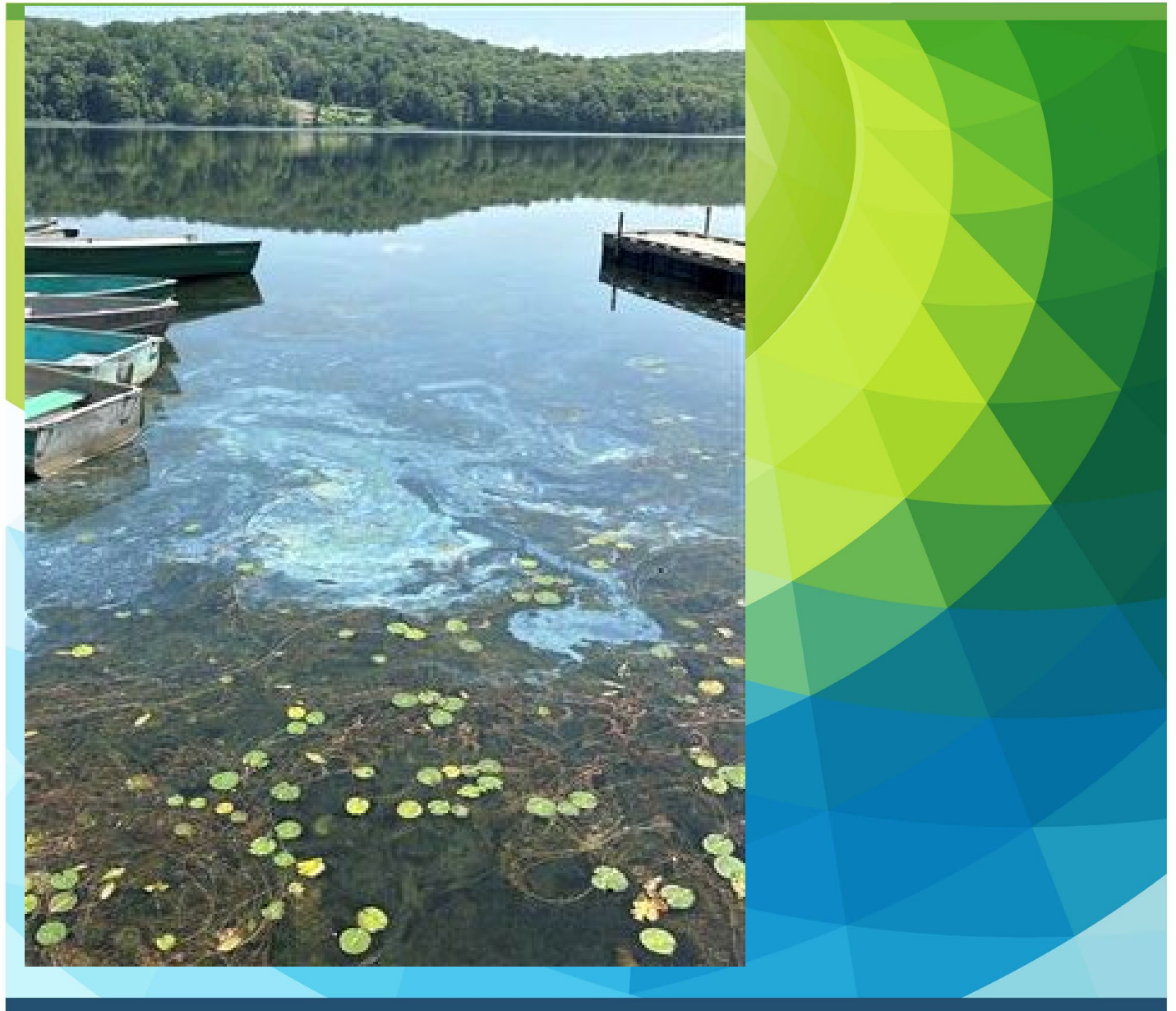


Cyanobacterial Harmful Algal Bloom (HAB) Freshwater Response

2025 Summary Report

Division of Water Monitoring, Standards and Pesticide Control



May 2026

**Cyanobacterial Harmful Algal Bloom (HAB)
Freshwater Recreational Response**

2025 Summary Report

New Jersey Department of Environmental Protection
Water Resource Management

Division of Water Monitoring, Standards and Pesticide Control

Bureau of Freshwater and Biological Monitoring

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May 2026

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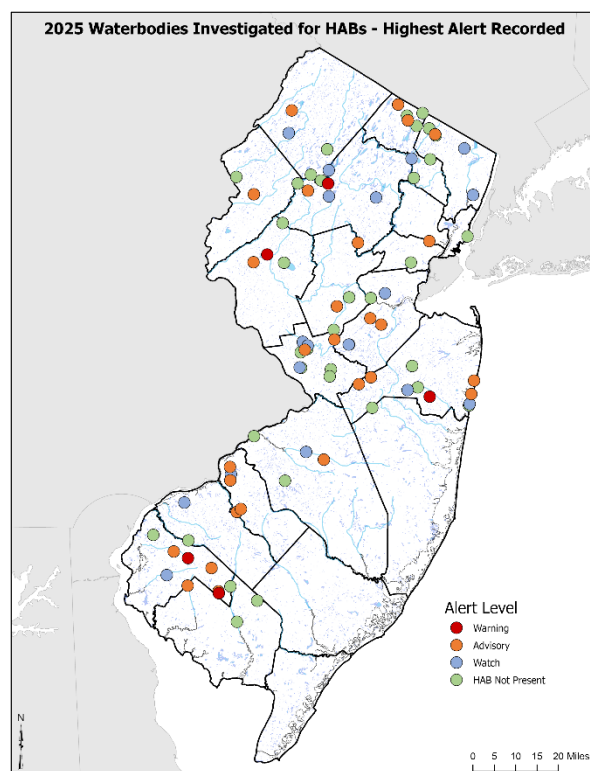
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Executive Summary

Figure 1. 2025 Waterbodies Investigated for HABs



In 2017, the NJDEP implemented a [Cyanobacterial Harmful Algal Bloom \(HAB\) Freshwater Recreational Response Strategy](#) (Response Strategy). The purpose of the Response Strategy is to provide a unified statewide approach to respond to cyanobacterial HABs in freshwater recreational waters, from public recreational bathing facilities to sources of drinking water, and to protect the public from risks associated with exposure to cyanobacteria and related toxins.

In 2020, a tiered public information and signage system was developed as an enhancement to the HAB Response Strategy. The alert tiers provide clear guidance on advisable recreational activities in waterbodies when a HAB is present. At the same time, [the DEP HAB Interactive Map Reporting and Communication System](#) was developed and is used to gather initial information on suspected HABs and to communicate data and alerts to

the public.

In 2025, DEP responded to reports of suspected HABs at 84 waterbodies. * Of these, 53 waterbodies had at least one site confirmed by laboratory analysis as having a HAB at or above a Watch Alert tier (>20,000 cells/ml and/ or toxins above thresholds).

*Data in this report reflect investigations of HABs reported to or discovered by DEP during routine monitoring. Other HAB events may have occurred and not reported to DEP.

The first confirmed HAB of the 2025 season was at Mountain Lake (Liberty) in early April (4/2/25). In total, 53 water bodies had confirmed HABs in 2025. Of those confirmed waterbodies, five waterbodies were deemed as Warning alerts (9%), 29 waterbodies were deemed as Advisory alerts (55%), and most waterbodies with confirmed HABs were at the Watch alert tier; 19 water bodies (or 35%) (Figure 2).

Figure 2. 2025 Confirmed HAB Summary

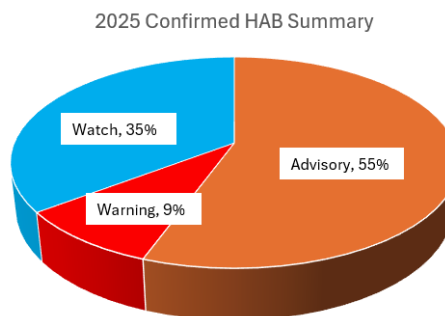
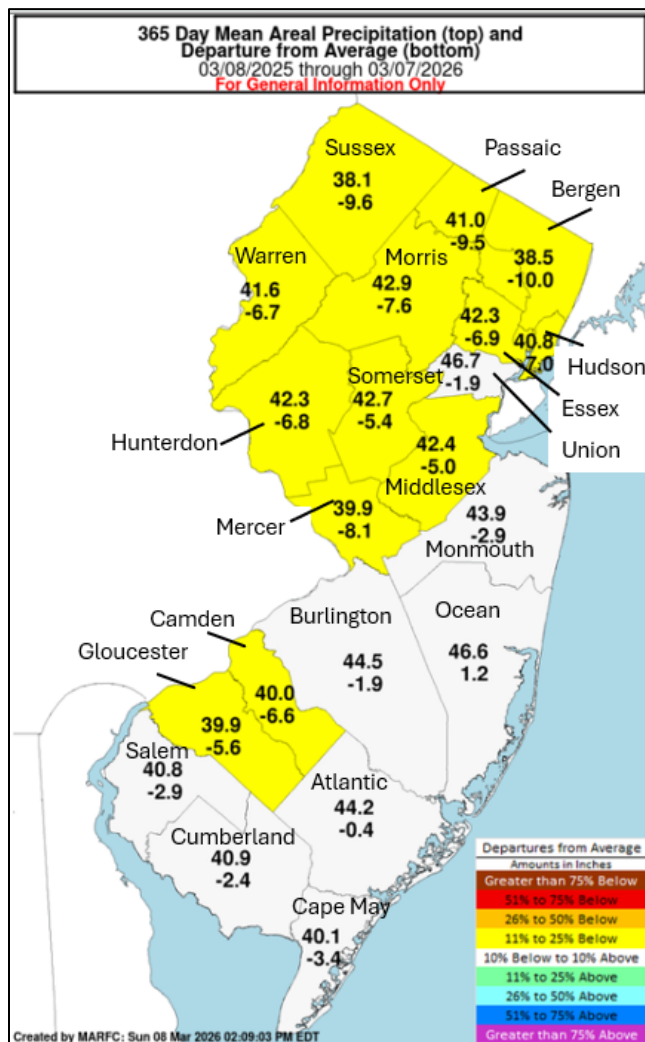


Figure 3. 365 Day Mean Precipitation and Departure from Average



Rainfall can contribute to the formation of HABs through various mechanisms: nutrient loading, increased turbidity, changes in water chemistry, transport of algal cells, and stratification disruption. Nutrient loading can occur during rain events which may carry phosphorus and nitrogen into water bodies which stimulates the rapid proliferation of HABs. Increased turbidity can reduce the amount of sunlight penetrating the bottom of the waterbody, creating low-light conditions which can encourage the growth of cyanobacteria.

During intense storms, algal cells can be transported and washed downstream to other water resources and can exacerbate the bloom intensity or duration. Rainfall events can disrupt thermal stratification by mixing surface and bottom waters causing resuspension of nutrient-rich bottom waters to the surface, further fueling algal growth.

Figure 3 depicts a map separated by county of the last 365 days of average rainfall. The top represents the number of inches of rainfall, while the bottom number indicates the departure from the average areal precipitation in inches. When both numbers are combined, this is the cumulative

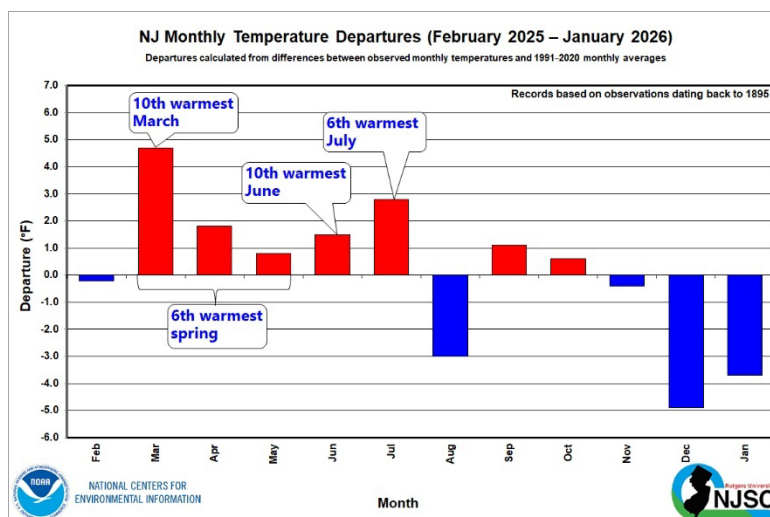
number (or measurement) of inches of rainfall the counties would have received under normal conditions. However, when the departure from the average number is divided by the cumulative rainfall amount, the percent will fall into one of the color-coded categories listed in the legend. Sussex, Warren, Morris, Passaic, Bergen, Hudson, Essex, Somerset, Hunterdon, Middlesex, Mercer, Camden and Gloucester counties were considered moderately drier than

normal and fell within the range of 11% to 25% below the average rainfall year to date. In those areas, some lakes continue to experience persistent HABs. Several other counties are below 10% or above 10% average rainfall in 2025.

Several months throughout 2025 were considered significantly drier than normal on record for New Jersey, exacerbating drought conditions across the state, increasing the likelihood in the occurrence of harmful algal blooms. January was one of the driest months on record, with only about 0.92 inches of precipitation statewide, making it the 3rd driest January in recorded history for New Jersey. On June 11th, the New Jersey Department of Environmental Protection lifted the Drought Warning that had begun in November 2024. Most counties returned to normal conditions except the south coastal region, which remained in a Drought Watch. Dry conditions continued into the summer season, with temporary improvement in precipitation and cooler temperatures observed in August, however, it was insufficient to alleviate the drought conditions fully. On October 1st, NJDEP expanded the Drought Watch to the entire state, marking a return of dry conditions statewide. By December 5th, NJDEP declared a statewide Drought Warning, reinstating the highest drought alert due to prolonged dry conditions.

Figure 4 displays a general summary of temperature conditions in 2025. Winter 2024–2025 was colder than the 1991–2020 climate average, and December 2024 also stood out with temperatures that were noticeably lower than those seen in recent years. Temperatures trended above normal beginning in March. March ranked as the 10th warmest on record, and the season in general ranked as the 6th warmest spring in New Jersey. This early warmth contributed to an accelerated start to the growing season statewide. August was cooler than average, with a negative departure, before temperatures drifted closer to normal in September and October.

Figure 4. NJ Monthly Temperature Departures (source: Rutgers)



The combination of low precipitation levels and elevated temperatures in 2025 significantly impacted New Jersey's environment and water resources, highlighting the state's vulnerability to climate variability, as well as susceptibility to potential of increased harmful algal bloom occurrence. Overall, 2025 was marked by a very warm spring and early summer, followed by a sharp transition to cooler-than-normal conditions late in the year.

Between 2018 and 2019, there was a 131% increase in reported waterbodies affected by HABs and an 84% increase in confirmed HABs (see Table 1). From 2019 to 2020, there was a 12% increase in reported HABs, and a 34% increase in confirmed HABs. However, from 2020 to 2021, both reported and confirmed HABs decreased significantly by less than 34%. This

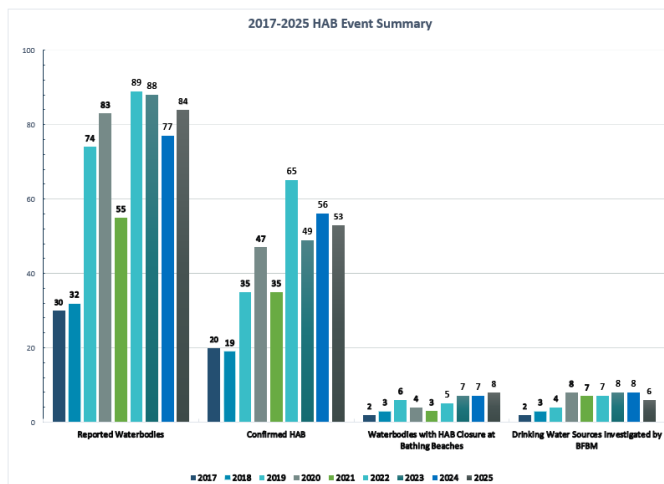
reduction in reporting may be attributed to the residual effects of the pandemic, such as increase in returning to the workforce.

From 2021 to 2022, reports increased by 61%, and confirmed HABs rose by 86%. In 2023, confirmed HABs decreased by 1% compared to the 2022 season. From 2023 to 2024, there was a 13% decrease in reported HABs, while confirmed HABs increased by 14%. Between 2024 and 2025, confirmed HABs increased by 9% and a decrease of 5% was observed in reported waterbodies. Displayed in Figure 5 is the percent change of increases and decreases of suspected vs confirmed HABs from year to year.

Year Comparisons	Reported Waterbodies	Confirmed HABs
2017 vs. 2018	+7%	-5%
2018 vs. 2019	+131%	+84%
2019 vs. 2020	+12%	+34%
2020 vs. 2021	-34%	-25%
2021 vs. 2022	+61%	+86%
2022 vs. 2023	-1%	-25%
2023 vs. 2024	-13%	+14%
2024 vs. 2025	+9%	-5%

Table 1. Harmful Algal Bloom Percent Occurrence

Figure 5. 2017-2025 HAB Event Summary



These 84 reports of suspected HABs represent a 9% increase from 2024 (Figure 5). This translated into a significant increase of waterbodies with confirmed HABs (Watch Alert or above) by 165% when compared to 2017, when the program began.

The number of confirmed HABs at drinking water sources decreased in comparison to 2024, while the number of waterbodies with beach closures also decreased to 6.

*Note that the Data shows the statewide occurrence of HABs in New Jersey have increased, and are recurring in

many waterbodies since 2017, when the DEP initiated monitoring per the [Response Strategy](#). Further evidence of continued HAB activity is the persistence of blooms in the winter months.

*Information of investigation at sources of drinking water are not to be used to interpret the safety of finished drinking water. The Department and the United State Environmental Protection Agency have established guidance levels and Health Advisory levels with respect to cyanotoxin detections in finished treated drinking water. The [DWSG has a guidance document](#) on when to issue public notification based on these levels.

The top 5 counties that samples were collected were in Morris, Passaic, Hunterdon, Mercer, and Monmouth county (Figure 6). Atlantic and Cape May counties did not have any samples collected this year. Note that some of these counties may contain larger lakes such as Lake Hopatcong, Greenwood Lake, etc. Among the samples counted for Passaic County, one was collected at Greenwood Lake, closer to the New York border, and has been incorporated into the overall sample count for the county. Therefore, more reports may occur at various locations throughout the lake resulting in more samples being collected to cover those different areas.

Figure 6. Samples Collected by County (2025)

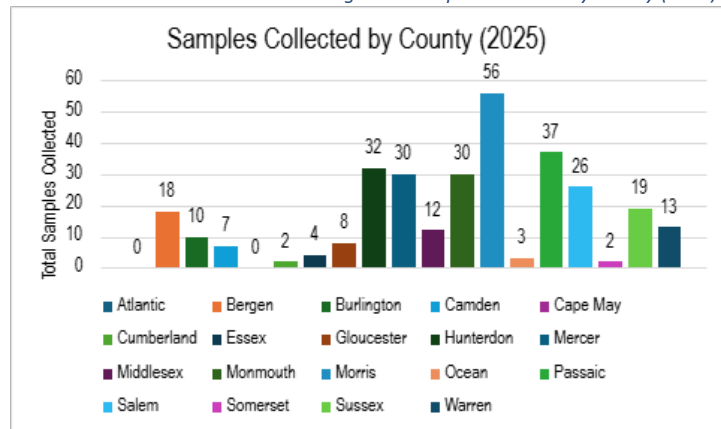


Table 2. Persistent HABs

Lake	Alert	Date
Elmer Lake	Advisory	11/18/25
Greenwood Lake – Lakeside Community Beach	Advisory	8/19/25
Greenwood Lake – Greenwood Lake Beach Club	Advisory	8/19/25
Greenwood Lake – Awosting Beach	Advisory	8/19/25
Greenwood Lake - Buoy	Advisory	8/19/25
Mountain Lake (Liberty) – Beach/dock	Advisory	12/1/25
Mountain Lake (Liberty) - Center of Lake	Advisory	12/1/25
Branta Pond	Advisory	12/10/25
Salem River – Commissioners Rd	Advisory	10/29/25
Sunset Lake	Advisory	11/24/25
Lake Owassa	Advisory	12/22/25
Roosevelt Park Pond	Advisory	8/15/25
Greenwood Lake (NY/NJ border)	Advisory	8/22/25
Spring Lake	Watch	11/24/25
Silver Lake	Watch	11/24/25
Fox Chase Pond	Watch	11/20/25
Budd Lake - Buoy	Watch	12/1/25
Budd Lake – boathouse ramp	Watch	12/1/25

As of the end of 2025, there were 10 waterbodies with at least one site persisting with a HAB Alert level of Watch or Advisory, i.e. the HAB had not dissipated (Table 2). Six waterbodies had a HAB Alert of Advisory and the remaining had a HAB Alert of Watch. This is an increase from 6 waterbodies in 2024. Waterbodies listed with an asterisk in Table 2 were posted as a Watch out of caution since BFBM crews could not visit these sites before the end of the year. A Winter Watch Alert was recommended for the 2025/2026 winter season at these waterbodies.

In addition to HAB response, the program has since implemented background monitoring, with the expansion of the continuous monitoring buoy network, [Home Page | NJDEP DWM&S Continuous Data Monitoring Program \(rutgers.edu\)](#). The continuous monitoring network provides valuable data at waterbodies where HABs have reoccurred. In addition to informing immediate HAB response actions, continuous data will be used by DEP to research water quality factors that may predict or contribute to HAB formation.

The Division of Water Monitoring, Standards and Pesticide Control (DWMSPC) and the New Jersey Sea Grant Consortium (NJS GC) recruited an Expert Team of lakes management and cyanobacterial HAB experts which developed the HAB Lake Management Guidance, [NJDEP | HABs | HAB Guidance](#), for the prevention and management of HABs and to provide technical advice on prevention and mitigation technologies.

Learn more about the [HAB expert team](#).

Introduction

In 2017, the NJDEP implemented a Cyanobacterial Harmful Algal Bloom (HAB) Freshwater Response Strategy (Response Strategy). The purpose of the Response Strategy is to provide a unified statewide approach to respond to suspected cyanobacterial HABs from freshwater recreational waters to sources of drinking water, and to protect the public from risks associated with exposure to cyanobacteria and related toxins. Although the primary focus of the Response Strategy is the protection of human health, it provides some information and recommendations regarding exposure and prevention of potential impacts to domestic animals (pets), livestock, and wildlife, as well.

The scope of the Response Strategy is for freshwater lakes, ponds, rivers and streams with potential public access, recreational use, public recreational bathing facilities as defined in N.J.A.C. 8:26, and sources of drinking water. These waterbodies may be owned or operated by state, county, municipal, federal, or private entities. As such, coordination of the investigation and response activities will vary depending on ownership and use.

It should be noted that while this Response Strategy is used to address sources of drinking water, it does not address the response for cyanotoxin detections in finished treated drinking water. The detection of cyanotoxins in finished treated drinking water is handled by the [Division of Water Supply & Geoscience](#) (DWSG) who has established guidance to best prevent, mitigate, and treat HABs/cyanotoxin as well as developing an emergency protocol for responding to and handling HAB/cyanotoxin events that affect a drinking water source/finished treated drinking water. However, the two Divisions work closely together along with the impacted water supplier during all stages of the incident to provide details and keep all relevant staff updated on the incident. Additional parties included in the coordination include but are not limited to, the Division of Water Enforcement, and other State agencies such as the New Jersey Department of Health, Board of Public Utilities, New Jersey Water Supply Authority, and New Jersey Department of Community Affairs, if appropriate.

Because the easiest way to deal with HABs/cyanotoxins is preventing them from happening, the DWSG also focuses on working with water systems to be better prepared for HAB/cyanotoxin events including creating and maintaining adequate Cyanotoxin Management Plans as well as convening a Drinking Water HAB Task Force to develop strategies and guidance. [The DWSG's HAB website](#) contains resources, tools, guidance, templates, and other useful information for water purveyors.

Since 2017, NJDEP has continued to enhance all aspects of its approaches including monitoring, testing, and communication/notification.

This report focuses on the response and monitoring performed in 2025. Data in this report reflect investigations of HABs reported to or discovered by DEP during routine monitoring. Other HAB events may have occurred but were not reported to DEP. For more information on other enhancements developed and implemented such as a data downloads, real-time telemetry buoys, and training videos, visit the [HAB Website](#).

Alert Tiers and Communication

In 2020, a tiered public information and signage system was developed as an enhancement to the HAB Strategy. The Alert tiers (Table 3) provide clear guidance on advisable recreational activities in impacted waterbodies, depending on levels of cyanobacteria and/or cyanotoxins present. Color-coded signs provide the public with current conditions and recommendations on which recreational activities are advisable and those that are not. The index makes it clear to the public that, in some instances, boating and related activities may still be suitable when lower levels of harmful algal blooms are detected.

While exposure to cyanobacterial cells that are not producing toxins can result in allergenic-like, flu-like and irritative effects, more serious health effects can result from exposure to cyanotoxins. Blooms may begin producing toxins at any time during an active HAB.

In developing the Alert tiers in 2020, DEP conducted an evaluation of NJ-specific HAB data to determine if there was a level of cyanobacterial cell density that is associated with an appreciable likelihood that a bloom will produce toxins at levels above the NJ toxin thresholds.

HAB ALERT LEVEL	CRITERIA	RECOMMENDATIONS
NONE	HAB report investigated and no HAB found	NONE
WATCH <i>Suspected or confirmed HAB with potential for allergenic and irritative health effects</i>	Suspected HAB based on visual assessment or screening test OR Lab confirmed cell counts between 20k – 40k cells/mL AND No known toxins above public health thresholds	Public Bathing Beaches Open (dependent upon local health authority evaluation and assessment) Waterbody Accessible: Use caution during primary contact (e.g. swimming) and secondary (e.g. non-contact boating) recreational activities Do not ingest water (people/pets/livestock) Do not consume fish
ALERT <i>Confirmed HAB that requires greater observation due to increasing potential for toxin production</i> PUBLIC BATHING BEACHES INCREASE MONITORING	Lab confirmed cell counts between 40k – 80k cells/mL AND No known toxins above public health threshold	WATCH remains in effect. Public Bathing Beaches Open (dependent upon local health authority evaluation and assessment) and should observe and report changing bloom conditions Waterbody Accessible: Use caution during primary contact (e.g. swimming) and secondary (e.g. non-contact boating) recreational activities Do not ingest water (people/pets/livestock) Do not consume fish
ADVISORY <i>Confirmed HAB with moderate risk of adverse health effects and increased potential for toxins above public health thresholds</i>	Lab testing for toxins exceeds public health thresholds OR Lab confirmed cell counts above 80K cells/mL OR Field measurement evidence indicating HAB present and above guidance thresholds (e.g. phycocyanin readings)	Public Bathing Beaches Closed Waterbody Remains Accessible: Avoid primary contact recreation (e.g. swimming) Use caution for secondary contact recreation (e.g. boating without water contact) Do not ingest water (people/pets/livestock) Do not consume fish
WARNING <i>Confirmed HAB with high risk of adverse health effects due to high toxin levels</i>	Toxin (microcystin) 20 - 2000 µg/l AND/OR Additional evidence, including, expanding bloom, increasing toxin levels (i.e. duration, spatial extent or negative human or animal health impacts) indicates that additional recommendations are warranted	Public Bathing Beaches Closed Waterbody Remains Accessible: Avoid primary contact recreation (e.g. swimming) May recommend against secondary contact recreation (e.g. boating without water contact) with additional evidence Do not ingest water (people/pets/livestock) Do not consume fish
DANGER <i>Confirmed HAB with very high risk of adverse health effects due to very high toxin levels</i>	Toxin (microcystin) > 2000 µg/l AND/OR Additional evidence, including, expanding bloom, increasing toxin levels (i.e. duration, spatial extent or negative human or animal health impacts) indicates that additional recommendations are warranted	Closure of Public Bathing Beaches Possible closure of all or portions of waterbody and possible restrictions access to shoreline. Avoid primary contact recreation (e.g. swimming) May recommend against secondary contact recreation with additional evidence Do not ingest water (people/pets/livestock) Do not consume fish

Table 3. HAB Alert Tiers

The HAB data were evaluated by analyzing the percentage of samples exceeding the NJ advisory guidance level for microcystins (the most common group of cyanotoxins) of 2 µg/L for various ranges of cyanobacteria cell counts. Cell count ranges were used to allow for enough samples for statistical analysis within each range. Each year this evaluation is performed using additional data collected from the previous season. Figure 7 displays the percentage of samples from 2017 to 2025 that exceeded the 2.0 µg/L microcystin threshold across different cyanobacterial cell count ranges. The highest exceedance rate occurs in both the 80,000 to <100,000 cells/mL range and the 100,000 to <500,000 cells/mL range, where over 30% of samples surpassed the threshold. In contrast lower cell count ranges (<40,000 cells/mL) show much lower exceedance rates, under 20%. When including 2025 results, the data continues to show an increase in the likelihood of toxin levels above the NJ guidelines when cell counts exceeded >80,000 cells/mL. (Figure 7)

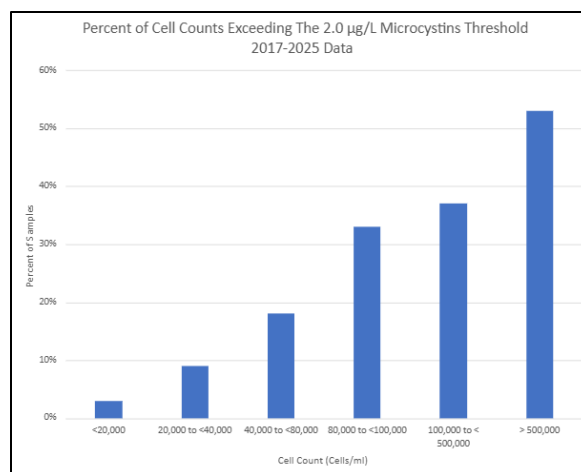


Figure 7. Percent of Cell Counts Exceeding The 2.0µg/L Microcystin Threshold

This graph shows the annual percentage distribution of samples that exceeded the 2.0 µg/L microcystin threshold, grouped by cyanobacterial cell count category <20,000 to >500,000 cells. Each year's bar represents 100% of samples that exceeded the threshold, and the stacked segments show what proportion those samples fell into each cell count range (Figure 8).

From 2017 to 2018 a large percentage of exceedance samples fell into the highest cell count category (>500,000 cells/mL), indicating more severe bloom conditions during exceedance events. In comparison to 2019 to 2021, the distribution shifts toward mid-range categories (40,000 – 500,000 cells/mL)

suggesting exceedances occurred more frequently at moderate cell densities. In 2022, A notable spike in the >500,000 cells/mL category, meaning a higher proportion of exceedance samples were associated with very high cell counts. From 2023 to 2025, a more balanced distribution, with a growing share of samples in the 100,000 – 500,000 cells/mL range and fewer dominated by the extremely high category. In general, the graph illustrates how severe cell densities were during toxin exceedance events each year. The severity of exceedance events varies year by year, with some years dominated by extremely high cell counts and others characterized by more moderate but still elevated densities.

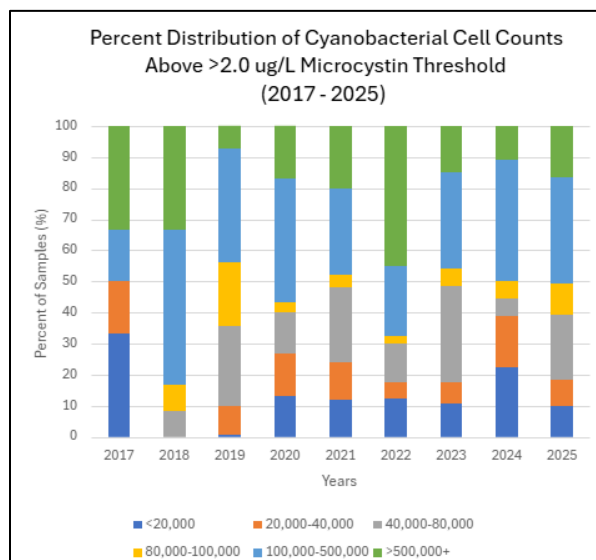


Figure 8. Percent Distribution of Cyanobacterial Cell Counts Above >2.0µg/L Microcystin Threshold (2017 – 2025)

A Winter Watch sign (Figure 9) was developed and implemented in 2021, to be used during the winter season. As shown by laboratory data, HABs may persist at some waterbodies or recur at other waterbodies. Day-to-day conditions may change and not reflect past Alert postings. The Winter Watch sign is intended to be used at these waterbodies where HABs have a likelihood of recurring during the winter. Because signs posted during the recreational season may not reflect current conditions, or the public may disregard signs they perceive as “old”, this new sign provides a fresh perspective during the off season to alert users to be cautious.

[The DEP HAB Interactive Map Reporting and Communication System](#) (Figure 10) was developed in 2020 and is used to gather initial information such as: location coordinates, photos, known recreational activities, and extent of the waterbody affected. This information is used to inform DEP to initiate appropriate response actions. After DEP completes the investigation of the suspected HAB, results and any recommendations for public alerts are communicated through the HAB System.

All Alert information and HAB data are accessible by clicking each point on the interactive map. The map reflects sampling results for suspected or confirmed HAB events reported to DEP.

In 2021, a new feature was added to the system that enables users to download all available data to date. Data downloads can be accessed here: [NJDEP Harmful Algal Bloom \(HAB\) Data Retrieval \(arcgis.com\)](#)

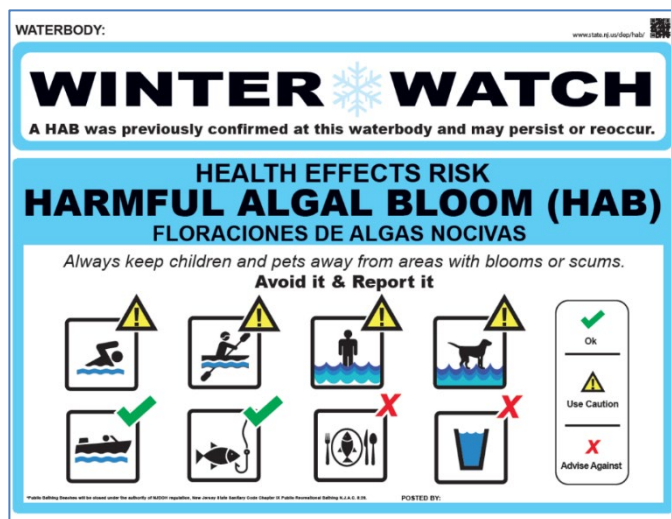


Figure 9. Winter Watch Sign

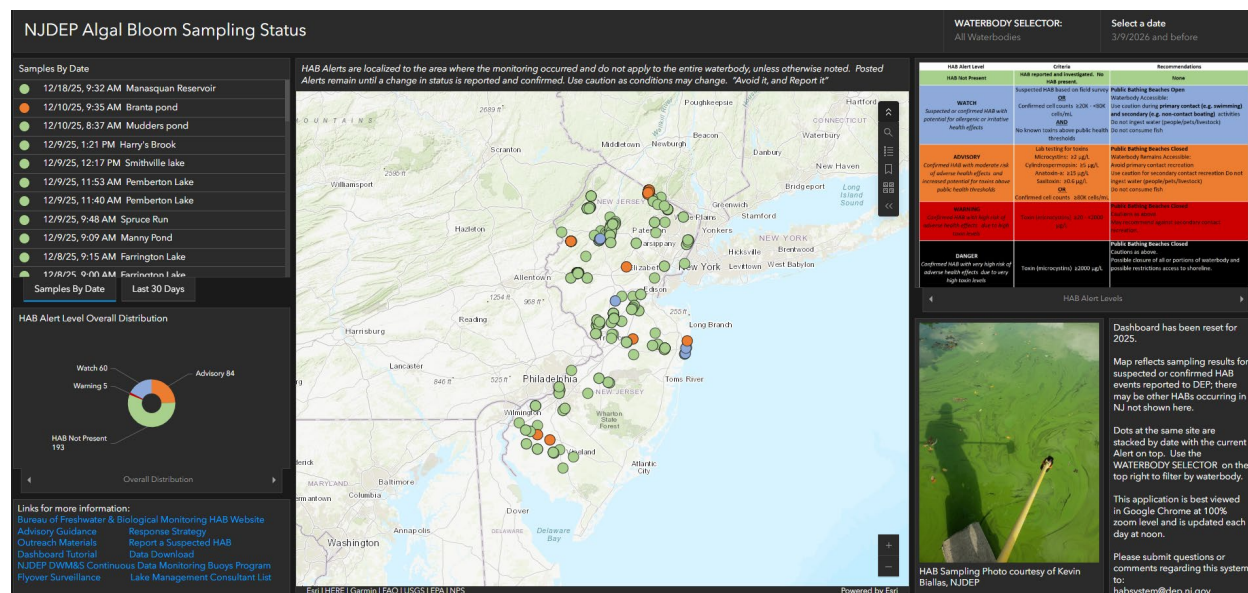


Figure 10. HAB Interactive Map.

Cyanobacterial Harmful Algal Bloom (HAB)

Freshwater Recreational Response

Procedures

Response

Response of suspected HABs is initiated in multiple ways both internally at DEP and outside. If trained DEP staff observe a suspected HAB while collecting water quality samples as part of the routine Ambient [Lakes Monitoring Network](#), [Rivers and Streams](#) routine monitoring networks, or other programs, a field survey, including sample collection (if necessary) is immediately conducted. Also, aircraft flight surveillance* and continuous monitoring buoy data are utilized by BFBM and DWMSPC designated staff to determine if a site survey and sampling is needed. Staff are deployed to areas of the waterbody, as shown by flight data or at buoy locations, where a HAB is suspected above NJ Guidance Thresholds for cell concentration.

Upon receipt of suspected HAB report from within DEP or the public, the BFBM HAB coordinator or designee assesses the information provided in the suspected HAB report, deploys staff, and/or coordinates with partners as necessary. The BFBM HAB coordinator also notifies the responsible agency designated for the waterbody, such as a State Park, Wildlife Management Area (WMA), or local health department. If the responsible agency has trained HAB sampling staff and proper sampling equipment and supplies, they may be requested to perform response activities.

Every effort is made to respond to reported suspected HABs as soon as possible, usually within one business day. In the event resources are limited, the monitoring will be prioritized based on risk to public health. Priority approaches are listed in Table 4.

*Flights are performed weekly weather permitting and described in Supporting Programs section.

Table 4. Response and Monitoring Priorities

1. Prioritization Response Approach for Lakes, Ponds, Reservoirs, Rivers & Streams including Delaware and Raritan Canal

a. Drinking Water Sources

Initial Response	Sampling Frequency	Duration/Season	Final Response
<i>Confirm ASAP</i>	<i>per Division of Water Supply & Geoscience direction</i>	<i>Year Round</i>	<i>Continue monitoring at predetermined frequency until clear or per Division of Water Supply & Geoscience direction</i>

b. Public Recreational Beaches (PRB) (in-season, out-of-season skip to c. Other Recreational Use) and Secondary Contact Recreational Waters

Initial Response	Sampling Frequency	Duration/Season	Final Response
<i>Confirm ASAP</i>	<i>Alert tier for bathing beaches only (see table 1) coordinate with partners on additional monitoring. Confirmed HAB Beach Closing – Sample when notified by partners that visual observations or phyco measurement indicates a change of HAB status.</i>	<i>May through September</i>	<i>After September 30 sample when notified HAB has visually subsided.</i>

c. Other Recreational Use - boating, fishing, public bathing beach (out of season), hunting, domestic animal use, wildlife

Initial Response	Sampling Frequency	Duration/Season	Final Response
<i>Confirm ASAP</i>	<i>Sample when notified by partners that visual observations or phyco measurement indicates a change of HAB status.</i>	<i>Recreational Season or Year Round if necessary</i>	<i>December. If HAB is still present or likely to reoccur, a “Winter Watch” alert is posted</i>

2. Approach for Private Lakes wholly on private property, Ditches, Canals, Stormwater Basins

Initial Response	Sampling Frequency	Duration/Season	Final Response
<i>Assess if there is public access (e.g. fishing or pet access in a private community). Contact owner. Sample on case-by-case basis.</i>	<i>As needed</i>	<i>As needed</i>	<i>When clear</i>

Field Survey

A field survey is performed to gather information following reports of suspected HABs. BFBM staff or partners record site coordinates, observations, take photos, and phycocyanin measurements. BFBM then determines if sampling is warranted. All survey and subsequent sampling information is recorded and submitted using the NJDEP HAB Interactive Map Reporting and Communication System.



Figure 11. Field Fluorometer

Phycocyanin is a pigment unique to cyanobacteria, therefore the presence of a high concentration of phycocyanin is an indicator of a cyanobacteria bloom. Handheld field fluorometers (Figure 11) measure the presence and relative concentration of phycocyanin and are used to qualitatively demonstrate whether cyanobacteria, if present, are in bloom densities. Phycocyanin measurements are used to approximate cell concentration and cannot predict toxin production, toxin levels, identify taxa present, nor quantify cell density directly. However, these measurements can be used as a screening tool for suspected HABs and to monitor the status of confirmed HABs.



Figure 12. Continuous multi-parameter meter.

BFBM uses three types of fluorometers: a handheld field meter, laboratory meter, and a YSI data sonde. The YSI data sonde is used for real time continuous monitoring in conjunction with telemetry buoys (Figure 12), but units can also be used for discreet measurements by samplers.

The DWMSPC has developed correlations between phycocyanin measurements and cell concentration. All New Jersey-specific data available, where both cell count and phycocyanin samples were analyzed, were used to statistically correlate these parameters. Note that the model of meter has different ranges and requires a separate correlation.

FluoroSense meters are used to screen waterbodies for suspected HABs and to monitor the status of a confirmed HAB. Phycocyanin and cell count data has been collected over the years in response to investigating HABs. As an additional preliminary screening tool, an evaluation of this data was conducted to determine a correlation between the two parameters. The graph below (Figure 13) displays the intensity of fluorescence emitted by phycocyanin ($\mu\text{g/L}$) using a

FluoroSense meter and cell counts (cells/ml) at the time of sampling and/or analysis.

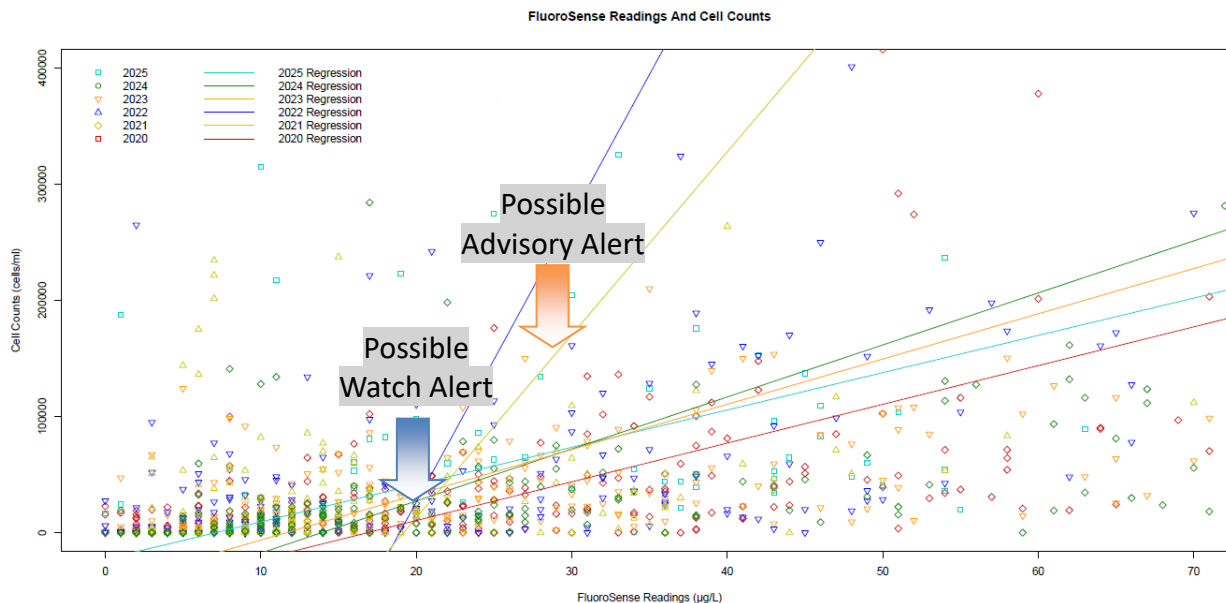


Figure 13. 2020 - 2025 Comparative Summary of FluoroSense and Cell Counts

From 2020 – 2025, the relationship between the fluorescence signal and the cell counts were generally linear, though the slope and the intercept of the regression line varied by year. This suggests that the fundamental relationship is consistent or maintained, however, slight protocol or calibration changes or biological variabilities may be influencing the exact relationship. In some years, the estimated slope was steeper, suggesting that a small change in the measurements was associated with a larger change in cell counts. In other years the slope was considered more moderate, showing that these shifts may reflect differences in characteristics of the microbial community or other sensitivity variables. Recognizing these shifts can guide or help in adjusting measurement protocols or in understanding inherent biological changes over time.

Looking at the coefficient of determination (R^2) serves as an indicator of how well the fluorescence measurements explain the variance in total cell counts in each year. Years where the R^2 was higher showed a stronger predictive capacity from the fluorescence measurements, while lower R^2 values indicated that additional factors may have been influential those years. As shown in the graph above, there are fluctuations in the linear regressions showing that the relationship is not entirely constant over time. Also taking into consideration the variability in the number of observations or samples being collected always plays a role. Years with larger datasets, shows regression results appeared to be more stable and reliable verses years with smaller datasets (smaller number of samples collected) the estimates were more susceptible to variability.

In the fall of 2025, additional analyses were conducted looking at over 2,776 samples (2020 – 2025) to determine generalized correlations between cell counts and FluoroSense readings. Note that the years 2017 and 2018 did not have FluoroSense data as the

equipment was not available during that time, and only partial information was available for 2019. Sensitivity tests were conducted on the dataset to screen additional skewness (including vs. not including non-existent data or data gaps) to enhance accuracy as much as possible. When calculating the mean (average), the results showed noticeable skewness, with concentrations in 10–15 µg/L range in relation to cell counts—an effect that was far less pronounced when using the median. Using the average may not be the best method to use when trying to determine the relationship between the Fluorosense readings vs. cell counts. By calculating the median and categorizing the measurements by cell count ranges provides a general representative number when screening for HABs. Note this is an estimated guidance based on the state-wide (NJ) data that NJDEP has compiled since 2020. Estimated readings based on data collected, generalized to the state of New Jersey. Individual water bodies may have different readings due to their unique characteristics. Table 5 shows the correlation range between Fluorosense readings vs. cell counts as well as other meters used.

Cell Count Ranges (Cells/ml)	Median FluoroSense Reading (µg/L)	Continuous & Discreet Meter (µg/L)	Estimated Lab Fluorometer (ppb)
20,000 to 39,999	20	1.15	33.00
40,000 to 79,999	29	1.87	45.90
80,000 to 99,999	43	3.29	71.50
>100,000	72	4.00	84.30

Table 5. Phycocyanin and Cell Count Correlation for All Meters Used.

Laboratory Analysis

Laboratory analysis is performed when it is confirmed that measurable cyanobacteria are present in a sample.

Toxin analysis is performed at the BFBM laboratory and uses an Enzyme-Linked Immunosorbent Assay (ELISA) method with Eurofins Abraxis brand test kits for cyanotoxin analysis of microcystins, anatoxin-a, cylindrospermopsin, and saxitoxin. Assays are performed using the Cyanotoxin Automated Analyzer System (CAAS) (Figure 14), Eurofins Abraxis brand, PN 475200S or equivalent Microtiter plate reader, capable of reading sample absorbance at 450 nm. Reporting levels for each toxin are adequate to accurately detect and quantify toxins below NJ Health Guidance.

Currently, EPA Standardized Analytical Method for Determining Total Microcystins utilizing the ELISA Method (EPA 546) is the only EPA-approved ELISA method for toxin analysis. Anatoxin-a, cylindrospermopsin, and saxitoxin are also analyzed using the ELISA methods. Procedures specific to these toxins follow the manufacturer's instructions for the kits and instrumentation.

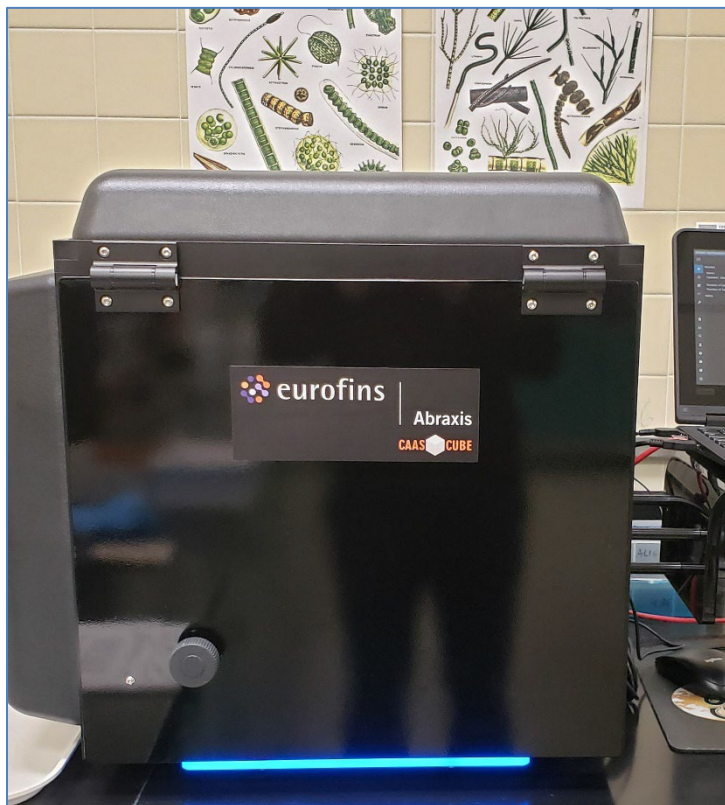


Figure 14. Cyanotoxin Automated Analyzer System (CAAS)

Cyanobacteria cell concentrations are determined in the BFBM laboratory by examining under a compound microscope using direct counts on a Hemocytometer. Standard phytoplankton identification guides are used for taxa identification. Cell counts are reported as cells/ml and all cyanobacteria taxa are identified. The dominant taxa, i.e., most abundant, is noted and posted with the data on the interactive map.

2025 Results and Discussion

Waterbody Summary

In 2025, BFBM responded to suspected HAB reports at 84 waterbodies. Of these, 53 waterbodies had at least one site, confirmed by laboratory analysis, as having a HAB at or above a Watch Alert level ($>20,000$ cells/ml and/or toxins above thresholds). A site was determined as not having a HAB when: field visual observations or phycocyanin measurements indicated no HAB was present and therefore a sample was not collected or; a sample was collected but lab analysis for cell count and toxins were below all thresholds. At each of the 84 waterbodies investigated for a suspected HAB, multiple sites may have been sampled, depending on extent of occurrences in the waterbody. In addition, sites may have been sampled many times over the season due to changing conditions and concerns. The Alert levels are intended for the immediate area where the HAB was confirmed through laboratory analysis, and the rest of the waterbody can be used for recreation with normal appropriate precautions. However, even when Alerts are posted, there may be other HABs occurring within that same waterbody or at other waterbodies, which have not yet been reported and confirmed. Therefore, recreators are advised to avoid anything that looks like a HAB and to report it to the DEP ("[Avoid It and Report It](#)"). Figure 15 shows a map of the waterbodies investigated in 2025 with the highest alert level recorded for that waterbody for the season.

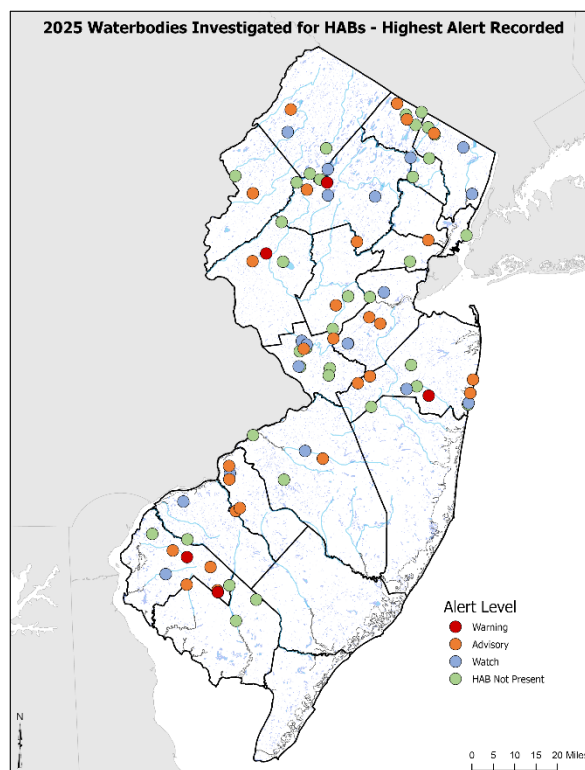


Figure 15. Map of 2025 Waterbodies Investigated.

Table 6. Waterbodies by County and Municipality

<u>Waterbody name</u>	<u>Highest Advisory Alert</u>	<u>County</u>	<u>Municipality</u>
Overpeck Creek	Watch	Bergen	Ridgefield Park Village
Darlington Lake Swimming Area	Advisory	Bergen	Mahwah Township
Darlington Upper Lake	HAB Not Present	Bergen	Mahwah Township
Woodcliff Lake	Watch	Bergen	Woodcliff Lake Borough
Macmillan Reservoir	HAB Not Present	Bergen	Mahwah Township
Pemberton Lake	Advisory	Burlington	Pemberton Township
White Birch Colony Club Pond	HAB Not Present	Burlington	Medford Township
Smithville Lake	Watch	Burlington	Eastampton Township
Amico Island Park Lake	HAB Not Present	Burlington	Delran Township
Cooper River	Advisory	Camden	Pennsauken Township
Haddon Lake	Advisory	Camden	Audubon Borough
Newton Lake	Watch	Camden	Collingswood Borough
Holly Run Lake	Advisory	Camden	Gloucester Township
Union Lake	HAB Not Present	Cumberland	Millville City
Bostwick Lake	Advisory	Cumberland	Hopewell Township
Memorial Park Pond	HAB Not Present	Cumberland	Vineland City
Weequahic Lake	Advisory	Essex	Newark City
Grenloch Lake	Advisory	Gloucester	Washington Township
Greenwich lake	Watch	Gloucester	Greenwich Township
Hudson River	HAB Not Present	Hudson	Jersey City
Spruce Run Reservoir	Warning	Hunterdon	Union Township
Round Valley Reservoir Swimming Impoundment	HAB Not Present	Hunterdon	Clinton Township
Manny Pond	Advisory	Hunterdon	Union Township
Old Canoe Pond	HAB Not Present	Hunterdon	Lebanon Township
Sylva Lake	HAB Not Present	Mercer	Ewing Township
Lake Ceva	Watch	Mercer	Ewing Township
Mercer Lake	HAB Not Present	Mercer	West Windsor Township
Dam Site 21	HAB Not Present	Mercer	Robbinsville Township
Willow Park	HAB Not Present	Mercer	Hopewell Township
Curlis Lake	HAB Not Present	Mercer	Hopewell Township
Wargo Pond	Watch	Mercer	Hopewell Township
Carnegie Lake	Advisory	Mercer	Princeton
Rosedale Lake	Advisory	Mercer	Hopewell Township
Honey Lake	Watch	Mercer	Hopewell Township
McCormack Lake	Watch	Middlesex	Plainsboro Township
D&R Canal	HAB Not Present	Middlesex	New Brunswick City
Farrington Lake	Advisory	Middlesex	East Brunswick Township
East Brunswick Municipal Pond	Advisory	Middlesex	East Brunswick Township
Papianni Lake	Watch	Middlesex	Edison Township

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<u>Waterbody name</u>	<u>Highest Advisory Alert</u>	<u>County</u>	<u>Municipality</u>
Lake McCormack	HAB Not Present	Middlesex	Plainsboro Township
Manasquan River	HAB Not Present	Monmouth	Howell Township
Wreck Pond	HAB Not Present	Monmouth	Spring Lake Borough
Assunpink Lake	Advisory	Monmouth	Upper Freehold Township
Lake Topanemus	HAB Not Present	Monmouth	Freehold Township
Turkey Swamp Park Lake	Watch	Monmouth	Freehold Township
Silver Lake	Advisory	Monmouth	Belmar Borough
Manasquan Reservoir	Warning	Monmouth	Howell Township
Millstone-Roosevelt Park Pond	Advisory	Monmouth	Millstone Township
Spring Lake	Watch	Monmouth	Spring Lake Borough
Sunset Lake	Advisory	Monmouth	Asbury Park City
Lake Musconetcong	HAB Not Present	Morris	Roxbury Township
Mudders Pond	Watch	Morris	Roxbury Township
Lake Rogerene	Warning	Morris	Mount Arlington Borough
Pompton River	HAB Not Present	Morris	Lincoln Park Borough
Budd Lake	Advisory	Morris	Mount Olive Township
Lake Parsipanny	Watch	Morris	Parsippany-Troy Hills Township
Prospertown Lake	HAB Not Present	Ocean	Jackson Township
Greenwood Lake	Advisory	Passaic	West Milford Township
Sheppards Lake	HAB Not Present	Passaic	Ringwood Borough
Monksville Reservoir	HAB Not Present	Passaic	Ringwood Borough
Haledon Reservoir	HAB Not Present	Passaic	North Haledon Borough
Ricker Drive Pond	Advisory	Passaic	Ringwood Borough
Ramapo River	Watch	Passaic	Pompton Lakes Borough
Erskine Lake	HAB Not Present	Passaic	Ringwood Borough
Game Creek	HAB Not Present	Salem	Carneys Point Township
Thundergust Lake	Warning	Salem	Pittsgrove Township
Parvin Lake	Advisory	Salem	Pittsgrove Township
Alloway Lake	Watch	Salem	Alloway Township
Memorial Lake	Advisory	Salem	Woodstown Borough
Harrisonville Lake	HAB Not Present	Salem	Pilesgrove Township
Green Branch	HAB Not Present	Salem	Pittsgrove Township
Salem River	Warning	Salem	Upper Pittsgrove Township
Elmer Lake	Advisory	Salem	Pittsgrove Township
Millstone River at Rt 518	HAB Not Present	Somerset	Franklin Township
Fox Chase Pond	Advisory	Somerset	Hillsborough Township
Powder Mill Pond	HAB Not Present	Somerset	Franklin Township
Branta Pond	Advisory	Somerset	Bernards Township
Lake Hopatcong	Watch	Sussex	Hopatcong Borough
Swartswood Lake	Watch	Sussex	Stillwater Township

<u>Waterbody name</u>	<u>Highest Advisory Alert</u>	<u>County</u>	<u>Municipality</u>
Lake Mohawk	HAB Not Present	Sussex	Sparta Township
Jefferson Lake	HAB Not Present	Sussex	Byram Township
Lake Owassa	Advisory	Sussex	Frankford Township
Jackson Pond	HAB Not Present	Union	Clark Township
Delaware Lake	HAB Not Present	Warren	Knowlton Township
Mountain Lake	Advisory	Warren	Liberty Township
Musconetcong River	HAB Not Present	Warren	Allamuchy Township

Note that occurrences of D&R Canal sites, Millstone River and larger lakes (i.e. Lake Hopatcong and Greenwood Lake) were consolidated into one entry per lake/river/canal in Table 6. Data and Alert Levels for all sites sampled can be found on the [DEP HAB Interactive Map Reporting and Communication System](#).

Public Recreational Bathing Beaches (PRB) and Drinking Water Sources

2025 Water bodies with beaches and in-season HAB closures:

- Darlington Lake – beach closures at 2 beaches
- Mountain Lake (Liberty)
- Lake Rogerene
- Budd Lake
- Lake Hopatcong – beach closures at 3 separate beaches
- Greenwood Lake – beach closures at 3 separate beaches
- Swartswood Lake
- Parvin Lake

Recreational Bathing Beach Confirmation

Listed above are the eight waterbodies with beaches and in-season HAB closures in comparison to Figure 6. Of the eight waterbodies, beach closures occurred at 13 individual bathing beaches. When a HAB was confirmed at Advisory/Beach Closure levels at Public Recreational Bathing Beaches (PRB) during the operating season of Memorial Day – Labor Day, the NJ Department of Health, Youth Camps/PRB Project Coordinator was immediately notified. The NJDOH PRB Coordinator then notified the appropriate local authority of the closure Alert and ensured onsite notices were posted. BFBM, or with the assistance of local authorities, monitored the status conditions of the HAB at these PRBs. The Strategy protocol recommends monitoring the HAB status at PRBs until bloom conditions dissipate to below Advisory/ Beach Closure levels, at which time samples are collected for laboratory confirmation analysis. Guidance in the Strategy further states that PRB closures should not be lifted until:

- With no phycocyanin field measurements - two (2) subsequent lab analyses were below cell count and toxin thresholds, or
- If phycocyanin measurements approximated cell counts below beach closing thresholds for consecutive days, then only one laboratory analysis with cell counts and toxin results below thresholds was necessary.

2025 Confirmed HABs at Recreational Waterbodies used as Drinking Water Sources*:

- Spruce Run Reservoir
- Manasquan Reservoir
- Farrington Lake
- Woodcliff Lake
- Ramapo River
- Oradell Reservoir

*Drinking Water Source Confirmation**

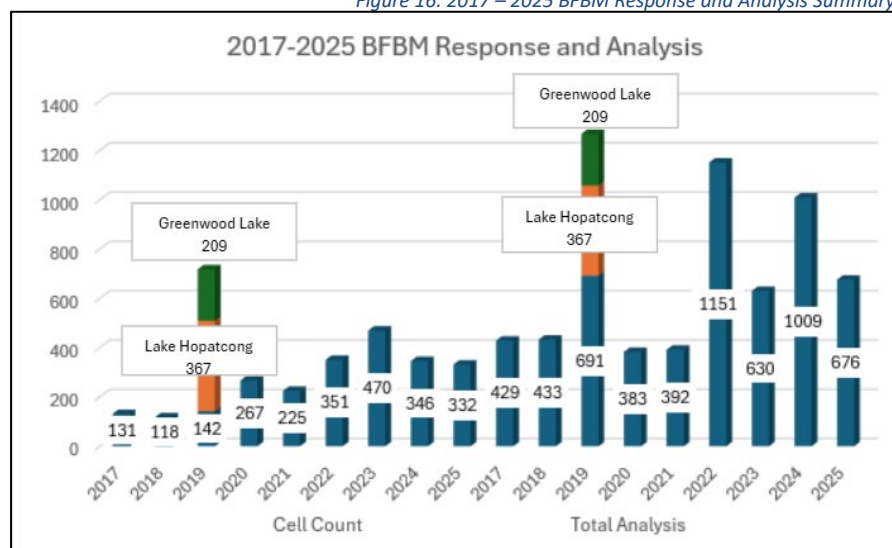
When a suspected HAB is reported at a possible drinking water source, BFBM immediately notifies the Division of Water Supply & Geoscience (DWSG) to confirm the location and possible use as a drinking source. If confirmed as a drinking water source, DWSG then informed the appropriate system operators who sampled their raw and finished water per their specific Cyanotoxin Management Plan. **This report summarizes drinking water sources investigated by BFBM that are also open to the public for recreation. Water suppliers may conduct their own investigations which are not reflected in this report.**

*The Advisories addressed in the document are for recreational public bathing beaches or sources of untreated drinking water and are not to be used to interpret the safety of finished drinking water. The Department and the United State Environmental Protection Agency have established guidance levels and Health Advisory levels with respect to cyanotoxin detections in finished treated drinking water. The [DWSG has a guidance document](#) on when to issue public notification based on these levels.

Laboratory Cell Count and Toxin Results

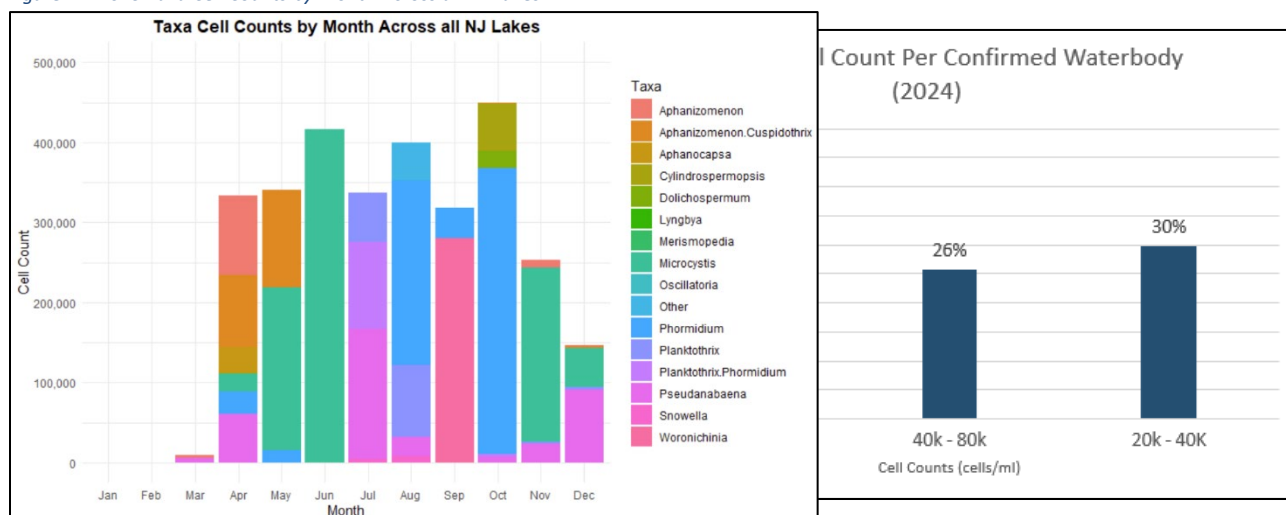
In 2025, laboratory analysis was consistent with the high number of reports investigated. There was a decrease (39%) in toxin analysis in 2025 due to the decrease in the number of confirmed HABs with microcystins over the 2.0 µg/l. When microcystins are present above 2.0 µg/l, it is protocol to analyze for the

Figure 16. 2017 – 2025 BFBM Response and Analysis Summary



other toxins as their likelihood of being present increases. Intensive surveys performed at Lake Hopatcong and Greenwood Lake in 2019 greatly increased the analysis performed that year. The higher numbers in 2017 and 2018 are due to samples collected at routine Ambient Lake Monitoring Network sites to develop toxin analysis capacity. Because routine lake sampling does not target active HABs, results were nearly all non-detect, or very low detection, unless a HAB was occurring at the time of sampling. Beginning in 2020, toxin sampling at Ambient Lake Monitoring Network sites was not performed unless an active HAB was visually observed or measured by field meters. As of 2020, analysis was focused strictly on response sampling. Therefore, toxin analysis in 2017-2019 was biased high compared to the number of sites investigated (Figure 16).

Figure 17. 2025 Taxa Cell Counts by Month Across all NJ Lakes



The graph depicts monthly cyanobacteria cell counts across all New Jersey lakes in 2025, grouped by taxonomic group, from January through December (Figure 17). Cell counts are minimal during winter months (January – March) and increase sharply in spring and early summer. Peak total abundances occur from June through October, with the highest levels observed in mid to late summer and early fall, frequently exceeding 400,000 cells/mL. Taxonomic composition exhibits pronounced seasonal variation: *Microcystis* predominates early and late summer, whereas *Phormidium* and *Planktothrix*-related taxa are most abundant in August and October. Additional taxa, such as *Woronichinia* and *Pseudanabaena*, attain peak abundances in mid to late summer, whereas several other groups persist at lower but relatively stable densities throughout the warm season. Overall, the data reveals a strong seasonal pattern in cyanobacterial blooms, with both abundance and taxonomic diversity peaking during the warm-water period from summer through early fall. Figure 18 summarizes the maximum cell count density at any given waterbody investigated during the 2025 season. In 2024, the majority (45%) of waterbodies with confirmed HABs had a peak cell count greater than 80,000 cells/ml, placing it in the Advisory Alert or higher category, in comparison to 2025 with cell counts

over 80k displayed a slight decreased to 35%.

An internal action level of 40,000 to 80,000 cells/ml initiates additional monitoring at bathing beaches only. This is to ensure the levels do not exceed the bathing beach closure threshold of 80,000 cells/ml and the proper Alert level is in place to protect bathers.

The highest recorded cell count concentration of 5,398,000 cells/ml, was at Salem River on 10/29/25.

However, the highest recorded toxin result of 2025 was at Thundergust Lake of 136.15 µg/L on 7/1/25 (Table 9). In the samples collected in 2025, as shown in Figure 19, the most common cyanobacteria taxa identified were *Aphanizomenon*, *Microcystis*, *Phormidium*, and *Aphanizomenon/Cuspidothrix*.

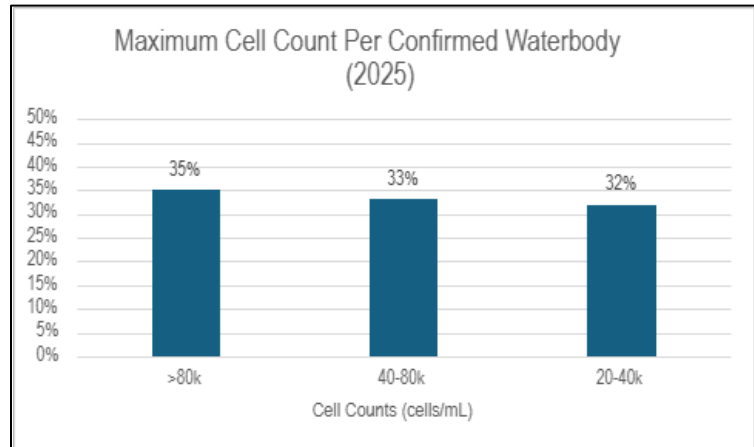
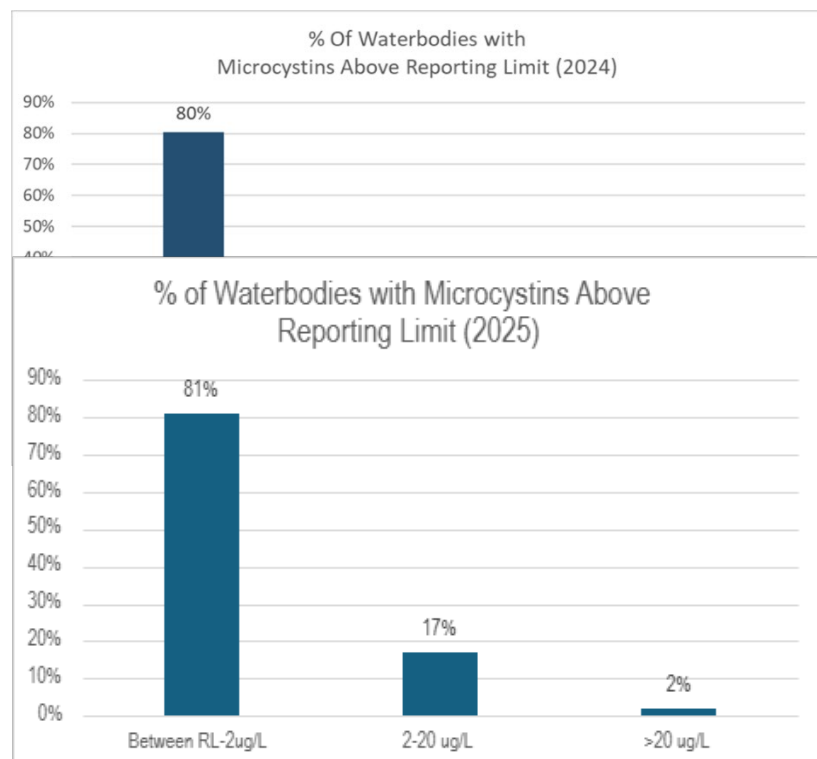


Figure 18. 2024/2025 Maximum Cell Count per Confirmed Waterbody

The majority of peak microcystins concentration at waterbodies with confirmed HABS, concentration at waterbodies with confirmed HABS, 81% were between the analysis Reporting Level (RL) of 0.15 µg/L and the Advisory Alert threshold of 2.0 µg/L. 17% were in the Advisory Alert category and 2% in the Warning Alert category. This is a slight increase in the toxin occurrence in the category of Advisory compared to the 2024 results. There was a 1% decrease in high toxin

Figure 19. 2024/2025 Microcystin Threshold Per Confirmed Waterbodies



occurrence in the categories of Warning Alerts in comparison to results from 2024 (Figure 19). The highest toxin concentrations occurred between the end of May to early October. The following waterbodies exceeded thresholds for the following toxins: Thundergust Lake for microcystins (136.15 µg/L), detections of saxitoxin (1.11 µg/L) were observed at Parvin Lake, detections of cylindrospermopsin (11.56 µg/L) in Greenwood Lake, and Fox Chase Pond with anatoxin (48.64 µg/L). Table 9 lists the waterbodies with the highest microcystins and cell count concentrations.

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Table 7. 2025 Cyanobacterial Taxa with MCT >2.0 ug/L

Cyanobacterial taxa observed in waterbodies with highest microcystins (MC) concentrations >2.0 µg/L during 2024 HAB responses in New Jersey.

[illegible]

Cyanobacteria recorded as present (+). Green highlight indicates the most dominate taxa in the sample. "+" indicate taxa are associated with the potential to produce microcystins.

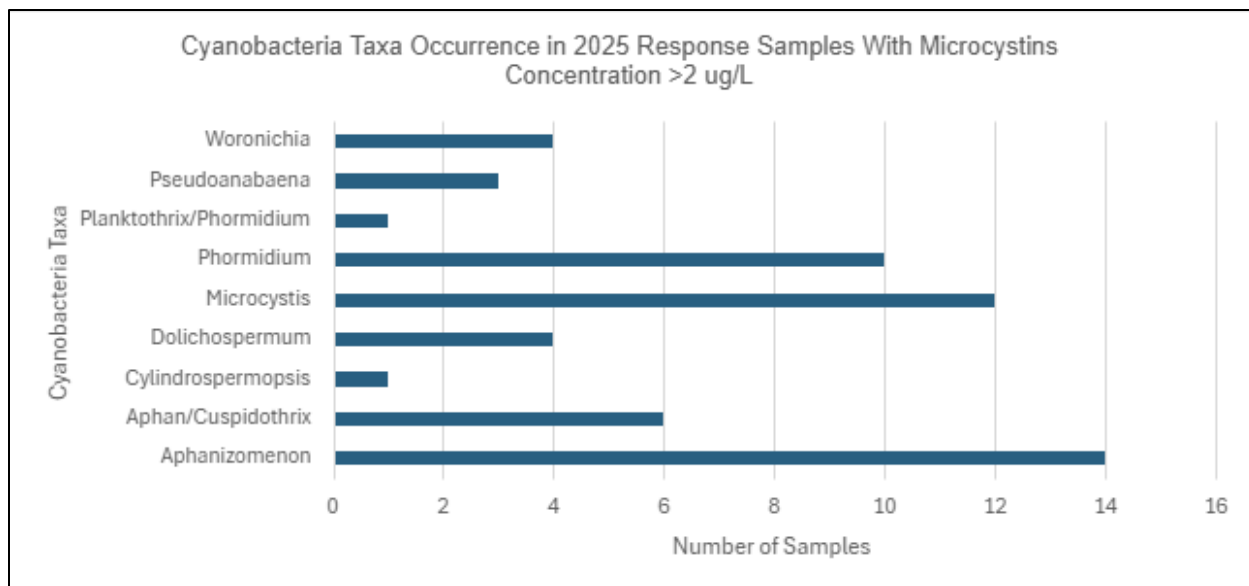


Figure 20. Cyanobacteria Taxa Occurrence in 2025 response samples

All samples were analyzed for microcystins. When microcystin results were above 2.0 µg/L, some samples were analyzed for the three other toxins if a waterbody had a significant recreational health risk and/ or cell concentrations were above 150,000 cells/ml. The toxins and their respective NJ recreational guidance thresholds are: anatoxin-a (15 µg/L), cylindrospermopsin (5 µg/L), and saxitoxin (0.6 µg/L). Anatoxin-a and cylindrospermopsin were detected at or above the lower detection limits of the tests in some samples, but none approached their recreational guidance threshold. Only cylindrospermopsin exceeded the 5 µg/L at one waterbody. Saxitoxin exceeded the 0.60 µg/L threshold at one waterbody. Anatoxin-a exceeded the 15 µg/L at one waterbody. Anatoxin-a was detected above the threshold at one waterbody which is an increase compared to 2024. Saxitoxin was also detected at two waterbodies above the NJ recreational threshold and is considered an increase. Detections of cylindrospermopsin decreased and was only observed at one waterbody in 2025. An internal literature review was performed to determine taxa with the potential to produce each toxin. The following percentages were detected above and below the threshold:

- Anatoxin-a (ATX) was detected in 30.0 % of these samples *Dolichospermum* sp., *Microcystis* sp., and *Woronichnia* sp. were associated with some of these blooms. All of these are known anatoxin-a producers.
- Cylindrospermopsin (CYL) was detected in 16.0 % of these samples. *Dolichospermum* sp., *Microcystis* sp., and *Woronichnia* sp. were associated with some of these blooms. Of these, only *Dolichospermum* sp. is a known cylindrospermopsin producer.
- Saxitoxin (STX) was detected in 25.4 % of these samples. *Aphanizomenon* sp., *Cuspidothrix* sp., *Dolichospermum* sp., *Microcystis* sp., *Lyngbya* sp., and *Woronichnia* sp. are associated with some

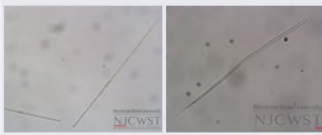
Most Common Dominant Cyanobacteria Taxa Found		
<i>Aphanizomenon</i>		
<i>Microcystis</i>		
<i>Phormidium</i>		
<i>Aphanizomenon/Cuspidothrix</i>		
Highest Cell Count	Salem River (5,398,000) cells/ml	

Figure 21. Most Common Dominant Taxa

of these blooms. Other than *Woronichinia sp.*, these are all known saxitoxin producers.

Table 8 provides additional information on highest cell count information. Table 9 lists the waterbodies with the highest Microcystin toxin concentrations.

Table 8. 2025 Highest Cell Counts

2025 Highest Cell Counts					
<u>Waterbody Name</u>	<u>Cell Count (cells/ml)</u>	<u>Total Cyano Genera</u>	<u>Predominant Taxa</u>	<u>Microcystins µg/L</u>	<u>Sample Date</u>
Salem River	5398000	3	<i>Woronichinia</i>	3.9	10/29/2025
Elmer Lake	1035000	2	<i>Microcystis</i>	1.54	8/22/2025
Salem River	990000		<i>Microcystis</i>	8.2	8/22/2025
Manasquan Reservoir	686875	3	<i>Microcystis</i>	41.74	10/7/2025
Memorial lake	509125	1	<i>Microcystis</i>	8.59	7/24/2025
Budd Lake	491500	3	<i>Microcystis</i>	0.65	8/20/2025
Spruce Run Reservoir	450625	4	<i>Woronichinia</i>	13.25	6/19/2025
Lake Rogerene	354000	3	<i>Microcystis</i>	104.5	8/19/2025
Spruce Run Reservoir	344625	3	<i>Woronichinia</i>	54.32	6/19/2025
Silver Lake	325000	1	<i>Cylindrospermopsis</i>	0.11	8/25/2025
Holly Run Lake	315000	2	<i>Microcystis</i>	2.35	9/16/2025
Millstone-Roosevelt Park Pond	274750	1	<i>Microcystis</i>	0.73	8/14/2025
Lake Owassa	236375	3	<i>Woronichinia</i>	0.18	9/29/2025
Manny Pond	223000	5	<i>Microcystis</i>	0.11	7/30/2025

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<u>Waterbody Name</u>	<u>Cell Count (cells/ml)</u>	<u>Total Cyano Genera</u>	<u>Predominant Taxa</u>	<u>Microcystins $\mu\text{g/L}$</u>	<u>Sample Date</u>
Carnegie Lake	222750	4	<i>Microcystis</i>	3.02	7/29/2025
Elmer Lake	217500	1	<i>Microcystis</i>	0.36	11/18/2025
Fox Chase Pond	205500	6	<i>Phormidium</i>	0.12	7/29/2025
Budd Lake	204125	2	<i>Microcystis</i>	0.78	5/28/2025
Grenloch Lake	187500	1	<i>Lyngbya</i>	0.1	8/18/2025
Farrington Lake	175875	3	<i>Woronichinia</i>	0.14	10/1/2025
Thundergust Lake	171125	1	<i>Microcystis</i>	136.15	7/1/2025
Rosedale Lake	153125	3	<i>Dolichospermum</i>	0.72	8/6/2025
Haddon Lake	153000	2	<i>Aphanizomenon</i>	0.43	8/12/2025
Branta Pond	147000	1	<i>Dolichospermum</i>	0.1	8/14/2025
Farrington Lake	137000	2	<i>Phormidium</i>	0.13	10/1/2025
Ricker Drive Pond	134000		<i>Microcystis</i>	0.11	8/14/2025
Cooper River	133625	2	<i>Dolichospermum</i>	0.46	7/22/2025
Pemberton Lake	125750	8	<i>Microcystis</i>	0.84	6/19/2025
Greenwood Lake	124125	5	<i>Aphanizomenon</i>	3.7	6/3/2025

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<u>Waterbody Name</u>	<u>Cell Count (cells/ml)</u>	<u>Total Cyano Genera</u>	<u>Predominant Taxa</u>	<u>Microcystins µg/L</u>	<u>Sample Date</u>
Spruce Run Reservoir	120500	4	<i>Dolichospermum</i>	3.55	6/19/2025
Farrington Lake	109375	3	<i>Phormidium</i>	0.22	10/1/2025
Rosedale Lake	106000	3	<i>Dolichospermum</i>	0.21	7/3/2025
Budd Lake	103625	2	<i>Microcystis</i>	0.9	10/1/2025
Spruce Run Reservoir	102875	2	<i>Woronichinia</i>	6.69	6/19/2025
Darlington Lower Lake	97500	1	<i>Microcystis</i>	0.07	8/12/2025
Greenwood Lake	96250	4	<i>Planktothrix/Phormidium</i>	2.32	7/2/2025
Spruce Run Reservoir	92375	4	<i>Microcystis</i>	10.7	6/19/2025
Salem River	91750	5	<i>Microcystis</i>	22.59	8/27/2025
Assunpink Lake	91000	3	<i>Dolichospermum</i>	0.27	7/7/2025
Greenwood Lake	89000	4	<i>Phormidium</i>	3.96	7/16/2025
Weequahic Lake	86000	4	<i>Aphanizomenon</i>	2.57	7/22/2025
Greenwood Lake	85750	5	<i>Aphanizomenon</i>	3.12	6/3/2025
Bostwick Lake	85750	3	<i>Microcystis</i>	0.28	8/7/2025

Cyanobacterial Harmful Algal Bloom (HAB) Freshwater 2025 Summary Report

<u>Waterbody Name</u>	<u>Cell Count (cells/ml)</u>	<u>Total Cyano Genera</u>	<u>Predominant Taxa</u>	<u>Microcystins µg/L</u>	<u>Sample Date</u>
Darlington Lake Swimming Area	83000	1	<i>Microcystis</i>	0.12	8/6/2025
Mountain Lake	82250	3	<i>Aphanizomenon</i>	3.58	5/28/2025
Mountain Lake	80750	3	<i>Aphanizomenon</i>	4.79	5/28/2025
Waterbody name	Cell Count (cells/mL)	Total Cyano Genera	<i>Predominant Taxa</i>	Microcystins (ug/L)	Sample Date
Salem River	5398000	3	<i>Woronichinia</i>	3.9	10/29/2025
Elmer Lake	1035000	2	<i>Microcystis</i>	1.54	8/22/2025
Salem River	990000		<i>Microcystis</i>	8.2	8/22/2025
Manasquan Reservoir	686875	3	<i>Microcystis</i>	41.74	10/7/2025
Memorial lake	509125	1	<i>Microcystis</i>	8.59	7/24/2025
Budd Lake	491500	3	<i>Microcystis</i>	0.65	8/20/2025
Spruce Run Reservoir	450625	4	<i>Woronichinia</i>	13.25	6/19/2025
Lake Rogerene	354000	3	<i>Microcystis</i>	104.5	8/19/2025
Spruce Run Reservoir	344625	3	<i>Woronichinia</i>	54.32	6/19/2025

Cyanobacterial Harmful Algal Bloom (HAB) Freshwater 2025 Summary Report

<u>Waterbody Name</u>	<u>Cell Count (cells/ml)</u>	<u>Total Cyano Genera</u>	<u>Predominant Taxa</u>	<u>Microcystins $\mu\text{g/L}$</u>	<u>Sample Date</u>
Silver Lake	325000	1	<i>Cylindrospermopsis</i>	0.11	8/25/2025
Holly Run Lake	315000	2	<i>Microcystis</i>	2.35	9/16/2025
Millstone-Roosevelt Park Pond	274750	1	<i>Microcystis</i>	0.73	8/14/2025
Lake Owassa	236375	3	<i>Woronichinia</i>	0.18	9/29/2025

In 2025, *Aphanizomenon* sp. (14 sites), *Microcystis* (12 sites), *Phormidium* (10), *Aphanizomenon/Cuspidothrix* (6 sites), *Dolichospermum* sp. (4 sites), *Woronichinia* (4 sites), *Pseudanabaena* (3 sites), *Planktothrix/Phormidium* (1 site) and *Cylindrospermopsis* (1 site) were the most frequently observed potentially toxigenic (PTOX) cyanobacteria taxa associated with the *Microcystis* sp. in most multispecies blooms with microcystins above 2.0 µg/L. (Table 9) These taxa, in addition to the *Microcystis* sp., are often associated with the production of microcystins.

Table 9. 2025 Highest Microcystin Toxin (Highest to lowest)

2025 Highest Microcystin Toxins (Highest to lowest)					
<u>Waterbody Name</u>	<u>Cell Count (cells/mL)</u>	<u>Total Cyano Genera</u>	<u>Predominant Taxa</u>	<u>Microcystins (µg/L)</u>	<u>Sample Date</u>
Thundergust Lake	171,125	1	<i>Microcystis</i>	136.15	7/1/2025
Lake Rogerene	354,000	3	<i>Microcystis</i>	104.5	8/19/2025
Spruce Run Reservoir	344,625	3	<i>Woronichinia</i>	54.32	6/19/2025
Manasquan Reservoir	68,6875	3	<i>Microcystis</i>	41.74	10/7/2025
Salem River	91,750	5	<i>Microcystis</i>	22.59	8/27/2025
Spruce Run Reservoir	450,625	4	<i>Woronichinia</i>	13.25	6/19/2025
Budd Lake	57,625	2	<i>Microcystis</i>	11.06	6/17/2025
Spruce Run Reservoir	92,375	4	<i>Microcystis</i>	10.7	6/19/2025
Mountain Lake	44,150	3	<i>Pseudanabaena</i>	10.23	12/1/2025
Greenwood Lake	40,375	3	<i>Aphanizomenon/Cuspidothrix</i>	10.09	8/19/2025
Memorial lake	509,125	1	<i>Microcystis</i>	8.59	7/24/2025
Salem River	990,000		<i>Microcystis</i>	8.2	8/22/2025
Mountain Lake	55,500	3	<i>Pseudanabaena</i>	8.02	4/2/2025
Greenwood Lake	54,000	5	<i>Aphanizomenon</i>	7.85	7/16/2025
Mountain Lake	38,250	2	<i>Pseudanabaena</i>	7.8	12/1/2025
Harry's Brook	48,500	2	<i>Dolichospermum</i>	7.65	6/9/2025
Greenwood Lake	39,000	3	<i>Phormidium</i>	6.99	8/6/2025
Greenwood Lake	33,375	4	<i>Phormidium</i>	6.89	8/6/2025
Spruce Run Reservoir	102,875	2	<i>Woronichinia</i>	6.69	6/19/2025
Greenwood Lake	50,375	5	<i>Aphanizomenon</i>	5.44	9/23/2025

Cyanobacterial Harmful Algal Bloom (HAB) Freshwater 2025 Summary Report

<u>Waterbody Name</u>	<u>Cell Count (cells/mL)</u>	<u>Total Cyano Genera</u>	<u>Predominant Taxa</u>	<u>Microcystins (ug/L)</u>	<u>Sample Date</u>
Greenwood Lake	53,000	4	<i>Phormidium</i>	5.1	7/16/2025
Mountain Lake	32,875	4	<i>Aphanizomenon</i>	4.88	6/23/2025
Mountain Lake	80,750	3	<i>Aphanizomenon</i>	4.79	5/28/2025
Thundergust Lake	47,875	2	<i>Microcystis</i>	4.3	7/16/2025
Greenwood Lake	46,500	2	<i>Aphanizomenon/Cuspidothrix</i>	4.16	6/11/2025
Greenwood Lake	47,000	5	<i>Phormidium</i>	4.12	8/19/2025
Greenwood Lake	53,375	2	<i>Aphanizomenon</i>	3.96	5/20/2025
Greenwood Lake	89,000	4	<i>Phormidium</i>	3.96	7/16/2025
Greenwood Lake	19,750	3	<i>Phormidium</i>	3.92	8/19/2025
Salem River	5,398,000	3	<i>Woronichinia</i>	3.9	10/29/2025
Greenwood Lake	64,500	5	<i>Aphanizomenon</i>	3.81	5/20/2025
Greenwood Lake	124,125	5	<i>Aphanizomenon</i>	3.7	6/3/2025
Greenwood Lake	35,875	4	<i>Phormidium</i>	3.69	8/19/2025
East Brunswick Municipal Pond	58,500	4	<i>Cylindrospermopsis</i>	3.67	10/3/2025
Mountain Lake	82,250	3	<i>Aphanizomenon</i>	3.58	5/28/2025
Spruce Run Reservoir	120,500	4	<i>Dolichospermum</i>	3.55	6/19/2025
Lake Rogerene	21,250	3	<i>Microcystis</i>	3.55	7/30/2025

Supporting Programs

As part of HAB response and monitoring, BFBM partners with several DEP and external partners. DEP's Division of Water Enforcement (C&E) aided in response screening and sampling. The State Park Service is also a significant partner providing assistance with response screening and sampling as well as posting Alerts when needed and monitoring the daily status of Park waterbodies.

NJ Forest Fire Service performs flight (Figure 22) surveillance at several larger Northern NJ lakes of concern. Visual observations are recorded as well as remote sensing of phycocyanin pigment. DWMSPC Bureau of Marine Water Monitoring developed a customized algorithm that can reliably detect and estimate phycocyanin concentrations in freshwaters through wavelength reflectance signatures.



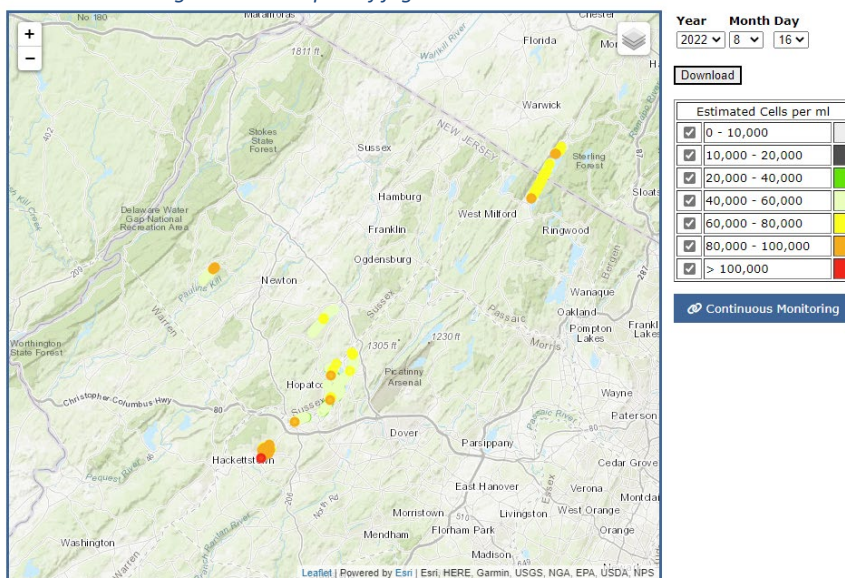
Figure 22. Forest Fire Service

These measurements are not used as a replacement for confirmation analysis, but as a screening and status monitoring tool to detect relative increases and decreases in phycocyanin pigment concentrations. When levels change significantly, i.e., indicate a change in Alert status, sampling staff are deployed for confirmation laboratory analysis.

As an enhancement in 2024, flight data was made available to the public at: [NJDEP New Jersey Department of Environmental Protection - Algal Bloom Remote Sensing \(rutgers.edu\)](https://njdep.state.nj.us/AlgalBloomRemoteSensing/) Flights were performed once per week during the recreational season at the following lakes (weather permitting):

Lake Hopatcong, Greenwood Lake, Musconetcong Lake, Budd Lake, Spruce Run Reservoir, Lake Mohawk, Swartwood Lake, and Round Valley Reservoir (non-HAB control lake). Other lakes were added as needed. Figure 23 shows examples of the flight data.

Figure 23. Examples of flight data.



The Bureau of Marine Water Monitoring has also assisted in developing a program using buoys equipped with continuous monitoring meters and real-time telemetry technology (Figure 24). Eleven (11) buoys were deployed at select waterbodies. The sites were chosen due to recreational and/or drinking water significance, repeated HAB occurrence, duration, and previous elevated levels of HABs at these waterbodies. The waterbodies had one or more remote monitoring devices to provide the best feasible coverage for HAB status monitoring & response:

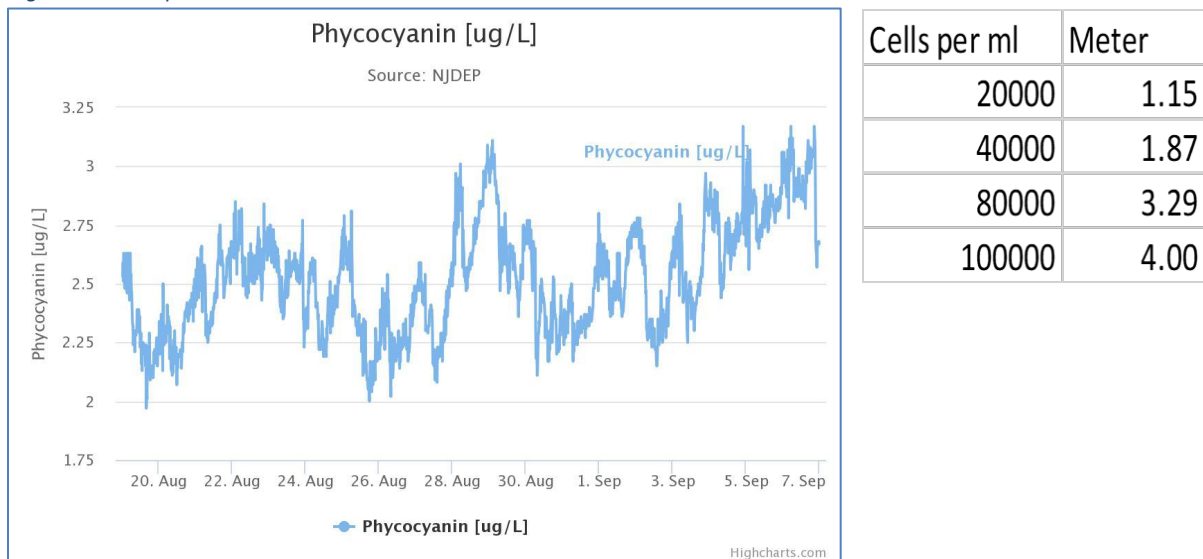
- Lake Hopatcong – 3 meters
- Spruce Run Reservoir– 1 meter
- Swartswood Lake – 1 meter
- Budd Lake – 1 meter
- Greenwood Lake – 1 meter
- Cozy Lake – 1 meter
- Greenwich Lake - 1 meter
- Manasquan Reservoir – 1 meter
- Carnegie Lake – 1 meter



Figure 24. Real-Time Continuous Buoy Program

These meters also measure other water quality parameters such as temperature, dissolved oxygen, and pH. As with phycocyanin measurements previously mentioned, this data is used for screening and status monitoring. Water quality data may be used to assess factors that may contribute to or characterize HAB production. Data can be downloaded or viewed in real time at the [NJDEP DWM&S Continuous Data Monitoring Program](#) website. Figure 25 shows an example of a downloadable graph with phycocyanin measurements at Greenwood Lake.

Figure 25. Example Real-Time Continuous Data Download



The program will continue in 2026. Based on data collected in 2025, buoys may be deployed at other waterbodies.

A field phycocyanin meter loan program was implemented in 2020 and expanded in 2022 from 12 to 40 meters respectively. No additional meters were distributed in 2025. These partners played a significant role in screening, status monitoring, and sampling. Meters were loaned to various internal and external partners:

- 10 -Water systems: AC MUA, North Brunswick WTP, Newton Twp, NJWSA (2), Sussex Borough Water, New Brunswick WA, City of Newark, Butler, NJ American North
- 5 -County/ local HDs: Sussex Co, Monmouth Co, Salem Co, Burlington Co, Hopatcong City Hall
- 2 -County/Local parks: Mercer Co, Hunterdon
- 7 -DEP Parks: North, Central, South, Swartswood, Wawayanda, Spruce Run (2)
- 5 -DEP C&E
- 12 – Commissions, Lake Associations, and Academia: Greenwood Lake Commission (2), Mountain Lake, Watershed Institute (3), Lake Hopatcong Foundation (2), Rock Lodge Lake Association, The College of New Jersey, Lake Owassa, Cranberry Lake, Branta Lake, Lake Hopatcong Commission

Conclusions

In 2025, there was a 9% increase in reported cases of suspected HABs compared to 2024. However, there was a 5% decrease in the number of waterbodies with confirmed HABs at the Watch Alert level or higher, relative to 2024. Analyzing reported waterbody data from 2017 to 2025 reveals a statewide increase of 95%. Furthermore, a comparison of confirmed HABs from 2017 to 2025 shows a rise of 113%. The data indicates that the statewide occurrence of HABs in New Jersey has escalated in numerous waterbodies since 2017, when the Department of Environmental Protection (DEP) began monitoring in accordance with the Response Strategy. In 2025, 28% of waterbodies with confirmed HABs had experienced confirmed HABs in previous years since the initiation of the HAB response in 2017 (Table 10).

Table 10. 2025 Confirmed HAB Waterbodies with Previous HABs

2025 Confirmed HAB Waterbodies with Previous HABs									
Waterbody Name	2017	2018	2019	2020	2021	2022	2023	2024	2025
Alloway Lake				X		X	X		X
Bellmawr Lake					X	X			
Branta Pond					X	X		X	X
Budd Lake	X	X	X	X	X	X	X	X	X
Canoe Pond					X	X			
Ceva Lake	X		X		X	X			
Cozy Lake				X		X	X		
Daretown Lake			X	X		X			
Duck Pond		X				X		X	
Elmer Lake			X			X		X	X
Farrington Lake				X	X	X		X	X
Greenwich Lake				X		X		X	X
Greenwood Lake			X	X	X	X	X	X	X
Haledon Reservoir				X		X			X
Lake Hopatcong			X	X	X	X	X	X	X
Lake Musconetcong			X	X	X	X		X	
Lake Sylvia	X		X		X	X			
Little Pond					X	X			
Manasquan Reservoir			X	X	X	X	X	X	X
Manny's Pond	X	X	X		X	X			X
Mountain Lake			X	X	X	X	X	X	X
Pemberton Lake	X		X	X	X	X	X	X	X
Pompton Lake				X		X			
Pompton River	X					X			
Quarry Pond				X		X			
Ramapo River				X		X			X
Rosedale Lake			X	X	X	X	X	X	X
Smithville Lake					X	X	X		X

2025 Confirmed HAB Waterbodies with Previous HABs									
Waterbody Name	2017	2018	2019	2020	2021	2022	2023	2024	2025
Spruce Run Reservoir			X	X	X	X	X	X	X
Sunset Lake			X	X		X		X	X
Swartswood Lake	X	X	X	X	X	X	X	X	X
Van Saun Mill Brook					X	X			
Woodcliff Lake Reservoir				X	X	X		X	X

Continued evidence of significant HAB activity is the persistence of blooms into the winter. Sampling and confirmation analysis for 2025 was completed in December and there were 10 waterbodies with at least one site with a HAB Alert level of Watch or above (Table 11). This is an increase from 6 waterbodies in 2024. Waterbodies listed with an asterisk in Table 11 were posted as a Watch out of caution since BFBM crews could not visit these sites before the end of the year. A Winter Watch Alert was recommended for the 2025/2026 winter season at these waterbodies.

Table 11. 2025 Persistent HABs

<u>Lake</u>	<u>Alert</u>	<u>Date</u>
Elmer Lake	Advisory	11/18/25
Greenwood Lake – Lakeside Community Beach	Advisory	8/19/25
Greenwood Lake – Greenwood Lake Beach Club	Advisory	8/19/25
Greenwood Lake – Awosting Beach	Advisory	8/19/25
Greenwood Lake - Buoy	Advisory	8/19/25
Mountain Lake (Liberty) – Beach/dock	Advisory	12/1/25
Mountain Lake (Liberty) - Center of Lake	Advisory	12/1/25
Branta Pond	Advisory	12/10/25
Salem River – Commissioners Rd	Advisory	10/29/25
Sunset Lake	Advisory	11/24/25
Lake Owassa	Advisory	12/22/25
Roosevelt Park Pond	Advisory	8/15/25
Greenwood Lake (NY/NJ border)	Advisory	8/22/25
Spring Lake	Watch	11/24/25
Silver Lake	Watch	11/24/25
Fox Chase Pond	Watch	11/20/25
Budd Lake - Buoy	Watch	12/1/25
Budd Lake – boathouse ramp	Watch	12/1/25

In previous years, cell concentrations above recreational guidance thresholds were the main reason for Alert postings. Toxin levels are usually associated with multispecies blooms where *Microcysts sp.* is present or dominant. Multi-species blooms (especially those containing *Aphanizomenon sp.*, *Cuspidothrix sp.*, *Dolichospermum sp.*, *Microcystis sp.*, and *Woronichnia sp.*) were present in samples where other toxins were detected by laboratory analysis. Therefore, it is recommended that the complete suite of toxins with recreational guidance thresholds in New Jersey be analyzed when these taxa are present in a sample.

The USEPA states (<https://www.epa.gov/cyanoHABs/causes-cyanoHABs>): “There is widespread agreement within the scientific community that the incidence of HABs is increasing both in the U.S. and worldwide. This recent increase in the occurrence of HABs has been attributed to increasing anthropogenic activities and their interaction with factors known to contribute to the growth of cyanobacterial blooms. Point sources (which may include discharges from municipal and industrial wastewater treatment plants, concentrated animal feeding operations (CAFOs), Municipal Separate Storm Sewer Systems (MS4s), stormwater associated with industrial activity, and other and non-point sources (which may include diffuse runoff from agricultural fields, roads and stormwater), may be high in nitrogen and phosphorus and can promote or cause excessive fertilization (eutrophication) of both flowing and non-flowing waters.”

The expansion of the continuous buoy network provides valuable data at waterbodies where HABs have reoccurred. In addition to informing immediate HAB response actions, continuous data will be used by DEP to research water quality factors that may predict or contribute to HAB formation.

In 2021, DWMSPC and the New Jersey Sea Grant Consortium (NJSGC) recruited a team of lakes management and cyanobacterial HAB experts to address the second component of the Governor’s HAB initiative focusing on enhancing scientific expertise and building the state’s capacity for HAB response. The HAB Expert Team’s primary objective is to provide guidance to DEP on HAB prevention, mitigation and management for NJ lakes and other waterbodies. As a culmination of their research, literature review, and data analysis, the ET developed the HAB Lake Management Guidance [NJDEP | HABs | HAB Guidance](#) which provides details to lake managers on how to develop a lake-specific action plan. The Guidance includes HAB prevention, mitigation, and management for NJ waterbodies that considers short-term and long-term action plan components. A virtual workshop was held in April 2023, which offered a comprehensive overview on the components of the Guidance. A recording of the Workshop and presentation materials are available on the HAB ET website [NJDEP | HABs | Expert Team](#). The Expert Team has successfully completed their charges.